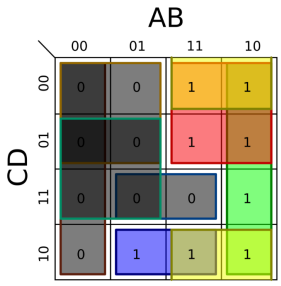
A 1.2.4 Truth Tables

Table of contents



A 1.2.4 Construct and analyse truth tables

Truth tables and Karnaugh maps are essential tools for unravelling the mysteries of logic circuits. Like detectives deciphering clues, we'll use these techniques to analyse circuit behaviour, simplify complex expressions, and optimise digital designs. Get ready to master the art of Boolean simplification!

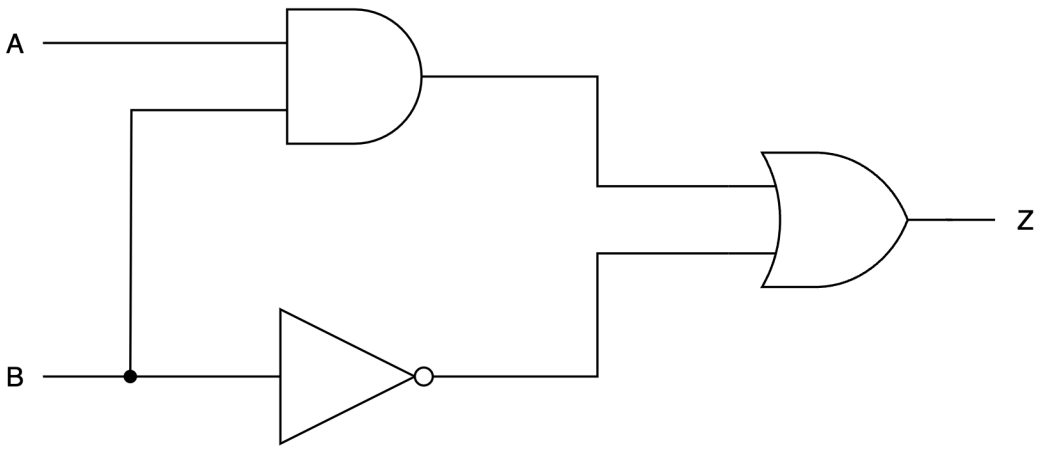
Last updated 25/10/2025

Key Questions

- How can truth tables be constructed and analysed for logic circuits?
- How are truth tables used to determine outputs from inputs and simplify logical expressions?
- How do Karnaugh maps assist in the simplification of output expressions?

Construct a truth table from a logic diagram

When given a circuit diagram, you need to break down the logic for each gate in the circuit. Consider the following diagram:



Determining Column Count:

- The number of columns in a truth table is directly linked to a circuit's inputs and logic gates.
- To calculate the total columns, simply add the number of inputs to the number of logic gates.
- For this logic diagram, there are two inputs and three gates, so we need 5 columns

Calculating Row Count:

- The number of rows in a truth table is determined by the number of input variables.
- The formula is 2 raised to the power of the number of inputs.
- So, a circuit with 4 inputs would have 2^4 = 16 rows. In this case, we have 2^2 which is 4 rows

A	B			Output Z
0	0			
0	1			
1	0			
1	1			

Before filling in the truth table, you need to determine the logic statement that defines the output for each gate: in addition to A and B, these will be the column headings.
The AND gate has inputs A and B, so its output can be defined as A.B
The NOT gate takes B, so the heading is $\overline{B}$
The OR gate takes the two previous outputs as inputs, so the heading for the final output is A.B + $\overline{B}$

A	B	A.B	$\overline{B}$	Output Z (A.B + $\overline{B}$)
0	0			
0	1			
1	0			
1	1			

Then we can complete the truth table based on these values.
We can then also create the logical expression from this truth table. The output is true (1) when:
$\overline{A}.\overline{B}$ + A.$\overline{B}$ + A.B
We can simplify this further by factoring out to
$\overline{B}$ + A.B
We will look at how to do this in more detail in A 1.2.5

A	B	A.B	$\overline{B}$	Output Z (A.B + $\overline{B}$)
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

How many rows are in a truth table?

Number of Input Combinations:

- 2 inputs: 4 possible combinations (2^2)
- 3 inputs: 8 possible combinations (2^3)
- n inputs: 2^n possible combinations

Creating a Truth Table for three inputs:

- Determine the number of input combinations based on the number of inputs. $2^3=8$
- Create columns for each input variable (e.g., A, B, C).
- Fill in the rows with binary sequences, starting from 000 and incrementing to 111.

When you draw a truth table, there is a useful technique for populating the rows and columns with 1 s and 0 s.

Step 1: Fill in the third row with alternating 0s and 1s

Inputs		
A	**B**	**C**
		0
		1
		0
		1
		0
		1
		0
		1

Step 2: Fill in the second row with pairs of alternating 0s and 1s

Inputs		
A	**B**	**C**
	0	0
	0	1
	1	0
	1	1

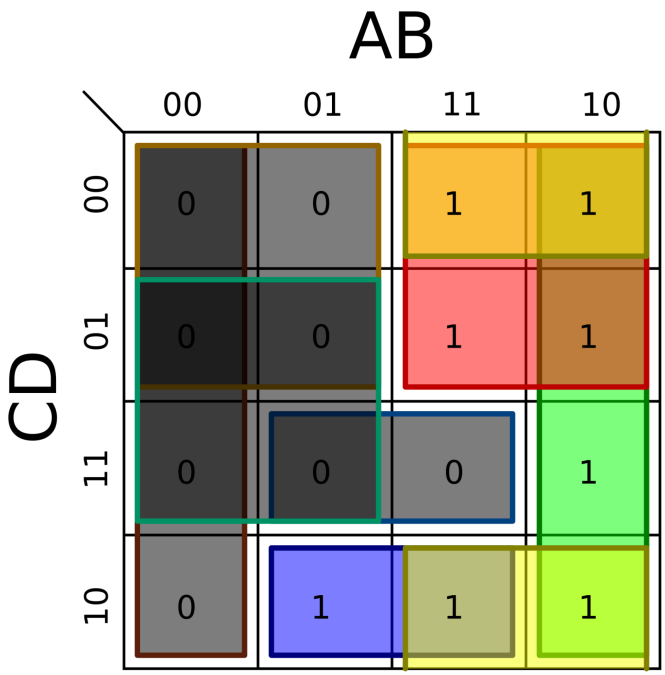
Inputs		
	0	0
	0	1
	1	0
	1	1

Step 3: Fill in the first row with quadruples of alternating 0s and 1s

Inputs		
A	**B**	**C**
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Karnaugh Maps

A Karnaugh map (KM or K-map) is a diagram that can be used to simplify a Boolean algebra expression.



f(A,B,C,D) = E(6,8,9,10,11,12,13,14)
F=AC'+AB'+BCD'+AD'
F=(A+B)(A+C)(B'+C'+D')(A+D')

From truth table to Karnaugh Map

Consider the truth table below.
This could be written as:
(A and not B) or (A and B)
Or:
$A.\overline{B} + A.B$
This is because the output (Z) is true for either of these input combinations.

Inputs		Outputs
A	B	Z
0	0	0
0	1	0
1	0	1
1	1	1

A Karnaugh map is a variation on a truth table, which is arranged such that:

- The number of output cells matches the number of rows in the truth table
- The possible input values for the variables (in the expression) form the row and column headings
- The output for each combination of input values is written at the row/column intersections

B \ A	0	1
0	0	1
1	0	1

$$A.\overline{B} + A.B$$

In the figure below, you can see that the inputs are A and B. The possible values for A have been used as column headings, and those for B have been used as row headings. At the intersection of the row and column is the output from the circuit for the given inputs.

B \ A	0	1
0	0	1
1	0	1

$$A.\overline{B} + A.B = A$$

Looking at this Karnaugh map, you can see that a group of 1 s is highlighted.
In a Karnaugh map, you circle any groups that have a bank of 1 s that is a power of 2 (i.e., when there are 1, 2, 4, 8 cells in the group, but not when there are 3, 5, 6, etc.).
There is only one group here, and it is in the column that shows the outputs where A = 1.
Because this is the only circled group, you know the equation simplifies to A.

Summary of the rules

- Groups must be horizontal or vertical; no diagonals are allowed
- The number of cells in each group must be a power of 2
- Groups should be as large as possible
- Every 1 in the map must be in at least one group
- You must create the smallest number of groups possible
- Groups can wrap around horizontally or vertically

Karnaugh Map to expression example 2

Looking at this Karnaugh map below, you can see 2 groups of 1s highlighted.
For the green group, the solution is $\overline{B}$ as this is the only constant value in the group.
For the yellow group, the solution is A, as this is the only constant value in the group.
Therefore the simplified expression is:
$A + \overline{B}$

B \ A	0	1
0	1	1
1	0	1

Three input Karnaugh Maps

For a Karnaugh map with three inputs, one row or column represents two of them.
Consider a map for three inputs A, B, C.
The map for this expression is shown to the right. In this map:

- There are four rows to represent the possible values of the inputs AB
- There are two columns to represent the possible values of the input C

AB \ C	0	1
00		
01		
11		
10		

Flexible Input Arrangement:

- The number of rows and columns in a Karnaugh map is not fixed.
- Inputs can be split across rows and columns in various ways. (BC, A etc.)

Sequential Input Value Changes:

- Only one input value should change between adjacent cells within a row or column.
- For example, if one cell has input values "01," the next cell could be "11" but not "10."

Non-Standard Row Order:

- The arrangement of rows in a Karnaugh map differs from a traditional truth table.
- This specific order is essential for grouping adjacent cells with similar output values.

Consider the expression:
(A.B) + (A.B.C)
Put a 1 in the cells where A.B is True
Put a 1 in the cells where A.B.C is True (in this case, the 1 has already been written)
Circle groups containing several 1 s that is a power of 2 (eg groups of 1, 2, 4, 8 cells)

AB \ C	0	1
00	0	0
01	0	0
11	1	1
10	0	0

This simplifies to:
A.B
This is done by the Absorption Law, which is one of the laws we look at later in A 1.2.5